Homework Assignment 4

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Question C-1 (d) Suppose we have the diophantine equation

$$x^2 + y^2 + z^2 = 8t - 1.$$

Reducing the equation by modulo 8, we have

$$\begin{aligned} x^2 + y^2 + z^2 &\equiv -1 \pmod 8 \\ &\equiv 7 \pmod 8. \end{aligned}$$

By division algorithm, any integer n is congruent to $k \pmod 8$, for some $k \in \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then, solving $n^2 \pmod 8$ for each n, we have

$$\begin{aligned} 0^2 &\equiv 0 \pmod 8, \\ 1^2 &\equiv 1 \pmod 8, \\ 2^2 &\equiv 4 \pmod 8, \\ 3^2 &\equiv 1 \pmod 8, \\ 4^2 &\equiv 0 \pmod 8, \\ 5^2 &\equiv 1 \pmod 8, \\ 6^2 &\equiv 4 \pmod 8, \text{ and} \\ 7^2 &\equiv 1 \pmod 8. \end{aligned}$$

Let $S = \{0, 1, 4\}$ be the set of all possible remainders from $n^2 \pmod 8$ for all integers n. Then, integers x^2, y^2, and z^2 are each congruent to $s \pmod 8$ for some $s \in S$.

Computing the sums of each unique combination of (x^2, y^2, z^2), we have

$$\begin{aligned} 0+0+0 &\equiv 0 \pmod 8, \\ 1+0+0 &\equiv 1 \pmod 8, \\ 1+1+0 &\equiv 2 \pmod 8, \\ 1+1+1 &\equiv 3 \pmod 8, \\ 4+0+0 &\equiv 4 \pmod 8, \\ 4+1+0 &\equiv 5 \pmod 8, \\ 4+1+1 &\equiv 6 \pmod 8, \\ 4+4+0 &\equiv 0 \pmod 8, \\ 4+4+1 &\equiv 1 \pmod 8, \\ \text{and } 4+4+4 &\equiv 4 \pmod 8. \end{aligned}$$

Since the sum $x^2 + y^2 + z^2 \pmod 8$ can only return values in $\{0, 1, 2, 3, 4, 5, 6\}$, and the right-hand side requires $7 \pmod 8$, the congruence $x^2+y^2+z^2 \equiv 7 \pmod 8$ is contradicted.

Therefore, we conclude the original diophantine equation $x^2 + y^2 + z^2 = 8t - 1$ has no solutions.

Question P-4

Let S be a set of 10 integers such that

$$S = \{x_1, x_2, ..., x_{10}\}$$

Define the sums $X_1, X_2, ..., X_{10}$ with

$$X_j = \sum_{i=1}^{j} x_i \quad \text{for each } j \in \{1, 2, ..., 10\}.$$

If there exists a j such that

$$X_j \equiv 0 \pmod{10},$$

then the sum of $\{x_1, ..., x_j\} \in S$ is divisible by 10. Thus, we are finished.

Otherwise, none of the sums $X_1, ..., X_{10}$ are congruent to 0 (mod 10).

By division algorithm, we know that the remainder r must satisfy $0 \leq r < n$, with $n = 10$ in this case. Since $r = 0$ is impossible, r must be in $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Thus, each of the sums $X_1, ..., X_{10}$ must be congruent to $r \pmod{10}$.

We choose these remaining remainders r to be the pigeonholes and the 10 sums $X_1, ..., X_{10}$ as the pigeons.

Then, by the Pigeonhole Principle, at least 2 of the sums from $X_1, ..., X_{10}$ will be congruent with each other under modulo 10.

We choose these 2 sums as X_i and X_j with $1 \leq i < j \leq 10$. Thus, we have that

$$X_j \equiv X_i \pmod{10}.$$

Subtracting both sides by X_i, we get

$$X_j - X_i \equiv 0 \pmod{10}.$$

But,

$$\begin{aligned} X_j - X_i &= (x_1 + ... + x_j) - (x_1 + ... + x_i) \\ &= x_{i+1} + ... + x_j. \end{aligned}$$

Thus, the sum of the non-empty subset $\{x_{i+1}, \ldots, x_j\}$ is divisible by 10.

Therefore, we conclude that for a set of ten integers X, there exists some non-empty subset such that the sum is divisible by 10.

Chapter 7, Question 30

Let there be integers a, b, and p, where p is prime.

For sake of contrapositive, we assume $p \nmid a$ and $p \nmid b$.

Since p is prime, its only positive divisors are 1 and p. Thus, because $p \nmid a$ and $p \nmid b$, it must be that gcd(p,a) = 1 and gcd(p,b) = 1.

Then by Bezout's Lemma, there exist integers x_1, x_2, y_1, y_2 such that

$$x_1 p + y_1 a = 1 \text{ and } x_2 p + y_2 b = 1.$$

Multiplying both equations together, we get

$$\begin{aligned} (x_1 p + y_1 a)(x_2 p + y_2 b) &= x_1 x_2 p^2 + x_1 y_2 pb + y_1 x_2 ap + y_1 y_2 ab \\ &= p(x_1 x_2 p + x_1 y_2 b + y_1 x_2 a) + ab(y_1 y_2) \\ &= 1. \end{aligned}$$

Let $w = x_1 x_2 p + x_1 y_2 b + y_1 x_2 a$ and $z = y_1 y_2$. Then, the equation reduces to

$$p(w) + ab(z) = 1, \text{ where } w \text{ and } z \text{ are integers.}$$

Thus, by Bezout's Lemma, $gcd(p, ab) = 1$. Then, $p \nmid ab$ is true, since otherwise $gcd(p, ab) = p$, which contradicts the previous statement that the gcd is 1. Thus, we have proved our contrapositive.

Therefore, we conclude that if $p \mid ab$, then $p \mid a$ or $p \mid b$.

Extra

We choose $a = 852$ and $b = 228$. Applying the Extended Euclidean Algorithm, we have

$$\begin{aligned} 852 &= 3(228) + 168, \\ 228 &= 1(168) + 60, \\ 168 &= 2(60) + 48, \\ 60 &= 1(48) + 12, \\ \text{and } 48 &= 4(12) + 0. \end{aligned}$$

This means $gcd(852, 228) = 12$. Next, in order to find two integers satisfying the conclusion of Bezout's Lemma, we compute

$$\begin{aligned} 12 &= 60 - 48 \\ &= 60 - (168 - 2(60)) \\ &= 3(60) - 168 \\ &= 3(228 - 168) - 168 \\ &= 3(228) - 4(168) \\ &= 3(228) - 4(852 - 3(228)) \\ &= 15(228) - 4(852). \end{aligned}$$

So $15(228) - 4(852) = 12$.

Now, we can write $\frac{a}{b}$ as the following continued fraction

$$\frac{a}{b} = 3 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1 + \frac{1}{4}}}}.$$

Let $\frac{p}{q}$ be the fraction resulting from erasing the final fraction within the continued fraction expansion of a/b. Giving us

$$\begin{aligned} \frac{p}{q} &= 3 + \frac{1}{1 + \frac{1}{2 + \frac{1}{1}}} \\ &= 3 + \frac{1}{1 + \frac{1}{3}} \\ &= 3 + \frac{3}{4} \\ &= \frac{15}{4}. \end{aligned}$$

Notice that $p = 15$ and $q = 4$ are exactly the absolute value of the coefficients from Bezout's Lemma.